

Henry Klatt

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Education

Ph.D. in Mathematics, The George Washington University 2022–present
Expected spring 2027. Advisor: Valentina Harizanov.
Dissertation working title: *Effective Nonstandard Algebra*.
Specialty examination: Passed with distinction, “Forcing in Computability Theory” (Mar. 22, 2024).
M.A. in Mathematics, The George Washington University 2024
B.S. in Mathematics, Iowa State University 2022
Magna cum laude.
Undergraduate thesis: *Language, Models, Ultraproducts, and the Hyperreals* (2021).
Advisor: Tim McNicholl.

Research Appointments and Fellowships

Research Fellowship, Hausdorff Research Institute for Mathematics Fall 2025
Trimester programme: “Definability, Decidability, and Computability,” Bonn, Germany.

Research Interests

Mathematical logic, primarily computability theory, with a focus on ultraproduct-like constructions in model theory and computability theory. Additional interests include knot theory, algorithmic learning theory, and number theory.

Preprints and Work in Preparation

Preprint

R. Dimitrov, V. Harizanov, **H. Klatt**, and K. Srinivasan. *On Cohesive Products of Fields* (2026).
Submitted to the *Journal of Symbolic Logic*. [arXiv:2604.09965](https://arxiv.org/abs/2604.09965).

Work in Preparation

V. Harizanov and **H. Klatt**. *HKSS Completeness of Quandles, Racks, and Keis*.
In preparation.

V. Harizanov, M. “Turbo” Ho, **H. Klatt**, M. Kukla, and K. Srinivasan. *Cohesive Powers of Abelian Groups*.
In preparation.

V. Harizanov, **H. Klatt**, and K. Srinivasan. *Generalized Cohesive Products*.
In preparation.

Teaching Experience

Instructor of Record, The George Washington University Summers 2023, 2025, 2026
Math and Politics. Independently responsible for course delivery, assessment, and grading.

Graduate Teaching Assistant, The George Washington University 2022–2026
Calculus I, II, and III; Math and Politics; Mathematical Ideas.

Talks

Invited Conference Talks

<i>Cohesive Products of Galois Extensions</i> Computability, Complexity, and Randomness 2026, University of Leeds.	Sept. 7–11, 2026
<i>Countable Ultraproducts</i> Logicón, Universidad Nacional Autónoma de México.	May 29, 2026
<i>Cohesive Products of Fields</i> Joint Mathematics Meetings, ASL Special Session on Computability and Its Applications, Washington, DC.	Jan. 5, 2026
<i>Cohesive Powers of Number Fields</i> AMS Eastern Sectional Meeting, hosted by the University of Connecticut, Hartford.	Apr. 6, 2025
<i>Quandles are Complicated</i> FRG Workshop on Definability, Decidability, and Computability, Harvard University.	Aug. 1, 2024
<i>A maximally complicated isomorphism problem</i> AMS Western Sectional Meeting, San Francisco State University.	May 12, 2024
<i>Quandles are Complicated</i> Knots in Washington, The George Washington University.	Dec. 10, 2023

Seminar Talks

<i>Cohesive Products of Galois Extensions</i> Logic Seminar, Iowa State University.	Sept. 22, 2026
<i>Cohesive Powers of Galois Extensions</i> Online Logic Seminar.	Apr. 23, 2026
<i>More on cohesive products of fields, Part II</i> University Seminar on Computability, Complexity, and Algebraic Structure, GW.	Feb. 11, 2026
<i>More on cohesive products of fields</i> University Seminar on Computability, Complexity, and Algebraic Structure, GW.	Feb. 4, 2026
<i>Introduction to Cohesive Powers</i> Junior Seminar, Hausdorff Research Institute for Mathematics.	Nov. 14, 2025
<i>Types, existential closures, and cohesive powers in the study of arithmetic</i> University Seminar on Computability, Complexity, and Algebraic Structure, GW.	Sept. 2025
<i>Why doesn't Chevalley's theorem solve the halting problem?</i> University Seminar on Computability, Complexity, and Algebraic Structure, GW.	Jan. 24, 2025
<i>Smaller Ultraproducts, part 2</i> University Seminar on Computability, Complexity, and Algebraic Structure, GW.	Oct. 16, 2024
<i>Smaller Ultraproducts: A Survey</i> University Seminar on Computability, Complexity, and Algebraic Structure, GW.	Oct. 2, 2024
<i>n-generic and more: how far can forcing take us?</i> University Seminar on Computability, Complexity, and Algebraic Structure, GW.	Mar. 27, 2024
<i>The isomorphism problem for quandles is complicated</i> Logic Seminar, The George Washington University.	Sept. 6, 2023
<i>Infinitesimal Calculus</i> Louise Hay Logic Seminar, University of Illinois at Chicago.	Oct. 3, 2022
<i>Ultrapower construction of the hyperreals</i> Logic Seminar, The George Washington University.	Sept. 15, 2022
<i>Some fruit from set-theoretic pluralism</i> Logic Seminar, Iowa State University.	Apr. 6, 2022
<i>How Effective is Forcing?</i> Logic Seminar, Iowa State University.	Nov. 30, 2021

Professional Service

Conference and Programme Organization

- Computability of Algebraic Structures** (scheduled) June 20–25, 2027
Lead organizer; BIRS Workshop, Casa Matemática Oaxaca (CMO).
- Special Session on Mathematical Logic** Oct. 3–4, 2026
Co-organizer; AMS Eastern Sectional Meeting, Washington, DC.
- Special Session: DDC@DC** Jan. 4–7, 2026
Co-organizer; Joint Mathematics Meetings, Washington, DC.
- Seminar on Computable Structure Theory** Fall 2025
Organizer; trimester programme: “Definability, Decidability, and Computability.”
Hausdorff Research Institute for Mathematics.
- Special Session on Computability, Complexity, and Algebraic Structure** Apr. 6–7, 2024
Co-organizer; AMS Spring Eastern Sectional Meeting, Howard University, Washington, DC.

Seminar Organization

- Logic Seminar**, The George Washington University 2022–present
Co-organizer.

Reading Group Organization

- Computable Structure Theory** Sept.–Dec. 2025
Co-organizer; Hausdorff Research Institute for Mathematics.
- Commutative Algebra** May–July 2025
Co-organizer; The George Washington University.

Honors and Research Support

- Travel Grants**, Association for Symbolic Logic 2024, 2025, 2026
- Summer Research Support** 2023, 2024
Supported through advisor Valentina Harizanov’s National Science Foundation grant.